

Supervisor Feedback Fixes — Summary Report

Regime-Aware ML Trading Project

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This document summarises the changes made in response to supervisor feedback. Each issue is listed with the corresponding fix, academic justification, and affected files. No drastic redesign was performed — all changes are targeted improvements within the existing architecture.

1. Motivation

Following the supervisor review of the thesis draft, eleven areas were identified for targeted improvement. These range from documentation fixes (signal localization, triple-barrier explanation) to methodological enhancements (feature normalization, additional validation methods, diagnostic metrics). The goal is to strengthen the academic rigor of the project without redesigning the existing pipeline.

2. Summary Table of Issues and Fixes

#	Issue	Fix	Files Modified
1	Signal localization unclear	Added per-detector documentation of how the signal is processed. Standardized signal names (yellow, red, green) and plot styling.	signals.py, localization.py, plot_utils.py
2	Triple-barrier explanation incomplete	Completed: entry=Close, TP=entry+2.0xATR, Generate the STP, (Section 6.1.1) =10, first barrier hit determined by STP.	Generate_STP.py, (Section 6.1.1)
3	Triangle latent bug (missing fix)	Fixed: _make_detail() call for desc_triangle_upper and desc_triangle_lower. Added atr_i parameter.	triangle.py, desc_triangle.py
4	Trend-leaking features in model	Refined: atr_14 (raw ATR), event_atr, obv_norm (normalized OBV), volume_norm (normalized volume). Generated features.py	atr_14.py, event_atr.py, obv_norm.py, volume_norm.py, features.py
5	No simple technical filter features	Added 8 binary features: bb_touch_upper/lower, indicator_cross_above/below, close_touch_sma50/200	bb_touch.py, indicator_cross.py, close_touch.py
6	No training-set confusion matrix	Added evaluate_model on train set. Results dict now has generate_train_test_split, model_fit, model_eval	evaluate_model.py, generate_train_test_split.py, model_fit.py, model_eval.py
7	No individual tree diagnostics	Added individual_tree_diagnostics(): per-tree test accuracy, error, time, size, ensemble improvement, histogram	individual_tree_diagnostics.py
8	No complementary k-fold CV	Added kfold_event_cv(): 5-fold contiguous disjoint folds, 80/20 train-test split in notebook	kfold_event_cv.py
9	Walk-forward CV lacks per-fold confusion matrices	Added per-fold confusion matrices and aggregate OMP to walk_forward_cv output.	walk_forward_cv.py
10	Notebooks and thesis text repeated all feature group tables, feature lists, generated the simplified notebooks 00/10	Removed all feature group tables, feature lists, generated the simplified notebooks 00/10	features.py, notebook_00.py, notebook_01.py, notebook_02.py, notebook_03.py, notebook_04.py, notebook_05.py, notebook_06.py, notebook_07.py, notebook_08.py, notebook_09.py, notebook_10.py
11	No summary PDF of changes	Created this document (supervisor_feedback_fixes_summary.pdf)	fixes_summary.pdf

3. Academic Justifications

3.1 Removing raw ATR and event_atr

Raw ATR(14) values scale linearly with SPY's price level: ATR was ~\$1 in 2010 and ~\$6 in 2025. Including it as a feature effectively encodes the calendar date, leaking temporal information that inflates model performance on chronological splits. The normalized version, **atr_ratio = ATR / Close**, measures volatility as a fraction of price and is stationary across different price regimes. event_atr has the same issue and was removed for the same reason.

3.2 Normalizing MACD features

The MACD line is defined as $EMA(12) - EMA(26)$. When SPY trades at \$100, a MACD value of 2.0 represents a 2% divergence; when SPY trades at \$600, the same 2.0 represents only 0.33%. Dividing by Close (**macd_norm = MACD / Close**) makes values comparable across different price regimes. The same normalization is applied to the signal line and histogram.

3.3 Removing obv_norm

On-balance volume (OBV) is a cumulative indicator: it starts at zero and accumulates signed volume over the entire history. Even after normalizing by recent volume, the cumulative sum trends upward or downward over time, encoding temporal position. This makes it a proxy for the calendar date rather than a useful trading feature.

3.4 Adding binary technical filter features

Eight binary (0/1) features encode simple, interpretable technical conditions: Bollinger Band touches flag extreme price positions; SMA(50)/SMA(200) crossovers capture the well-known golden cross/death cross signals; Close proximity to key SMAs flags potential support/resistance tests; RSI extremes flag overbought/oversold conditions. These add low-complexity interpretable signals that any practitioner would recognize, without introducing new detectors or increasing model complexity.

3.5 Training-set confusion matrices

Evaluating on the training set is standard practice for diagnosing overfitting (Hastie, Tibshirani & Friedman, 2009). A model that achieves near-perfect training accuracy but poor test accuracy is memorizing noise. The train-test accuracy gap quantifies this risk. Including train confusion matrices alongside val/test matrices provides a complete picture of model behaviour across all data partitions.

3.6 Individual tree diagnostics

Ensemble methods derive their power from aggregating diverse weak learners. If the ensemble accuracy is only marginally better than the mean individual tree accuracy, the voting mechanism adds little value. If a few strong trees dominate while most are weak, the ensemble is fragile. Reporting per-tree accuracy statistics (mean, std, min, max) and ensemble improvement quantifies ensemble stability and helps determine whether the ensemble size (200 trees) is sufficient.

3.7 Five-fold event-level CV

With only ~137 labeled events and a 60/20/20 split producing ~28 test events, single-split metrics have high variance. K-fold cross-validation reduces this variance by averaging over multiple test folds (Kohavi, 1995). Using contiguous, pairwise disjoint folds preserves some temporal structure while maximizing each fold's effective test size. This is explicitly presented as a **complementary** diagnostic, not a replacement for walk-forward CV which respects strict temporal ordering.

4. Before / After Summary

Aspect	Before	After
Feature count	~49 (with atr_14, event_atr, obv_norm, raw MACD)	MACD with atr_ratio, normalized MACD, 8 binary filters, no leakers)
MACD features	Raw: macd, macd_signal, macd_hist	Normalized: macd_norm, macd_signal_norm, macd_hist_norm
Signal localization docs	Implicit in code	Explicit per-detector documentation in docstrings and thesis
Triple-barrier explanation	Brief paragraph	Explicit bullet list: entry, TP, SL, max_holding, label rule
Triangle bug	Missing atr_i in desc_triangle_upper_test	Fixed: atr_i passed correctly
Train confusion matrices	Not included	Train/val/test CMs with train-test gap table
Individual tree diagnostics	Not included	Per-tree accuracy stats + histogram + ensemble improvement
K-fold CV	Not included	5-fold event-level CV with mean/std accuracy and F1
Walk-forward per-fold CMs	Not included	Per-fold and aggregate confusion matrices
Validation methods	Chrono split + walk-forward	Chrono split + walk-forward + 5-fold event CV + train CMs

5. Files Modified

File	Changes
src/patterns/triangles.py	Fixed missing atr_i bug; added signal localization docstring
src/patterns/channels.py	Added signal localization docstring
src/patterns/support_resistance.py	Added signal localization docstring
src/patterns/multiple_tops_bottoms.py	Added signal localization docstring
src/features/indicators.py	Removed atr_14, obv_norm from compute_all_indicators; normalized MACD; added binary_technical_
src/features/build_features.py	Removed event_atr from feature matrix
src/models/train.py	Added individual_tree_diagnostics(), kfold_event_cv(); added train-set evaluation; added per-fold CMs
reports/generate_thesis.py	Updated feature tables, signal localization section, triple-barrier explanation, train CMs, tree diagnostics
notebooks/09_feature_engineering.ipynb	Updated conclusion, feature descriptions
notebooks/10_model_training.ipynb	Updated imports, feature description; added cells for train CMs, tree diagnostics, 5-fold CV
reports/generate_supervisor_feedback_summary.py	Renamed (this report)

6. Remaining Limitations

- **Small sample size** remains the primary constraint (~137 events). No amount of cross-validation can overcome limited data.
- **No real Alpha Vantage validation** — the comparison still uses simulated noise, not a real API key.
- **No hyperparameter optimization** — max_depth=8 and n_estimators=200 are fixed defaults.
- **5-fold event-level CV does not respect temporal ordering** — it is a complementary diagnostic, not a replacement for walk-forward CV.
- **No regime conditioning** (HMM) or backtesting with transaction costs.

These limitations are honestly documented in the thesis and do not invalidate the methodological improvements made in this round of fixes.